CSC 309: Artificial Intelligence

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2025/2026 Session

Week 3 : Constraint Satisfaction Problems (CSPs)

Week 3: Outline

- Constraint Satisfaction Problems (CSPs)
- Backtracking Search for CSPs
- Local Search for CSPs

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Constraint Satisfaction Problems (CSPs)

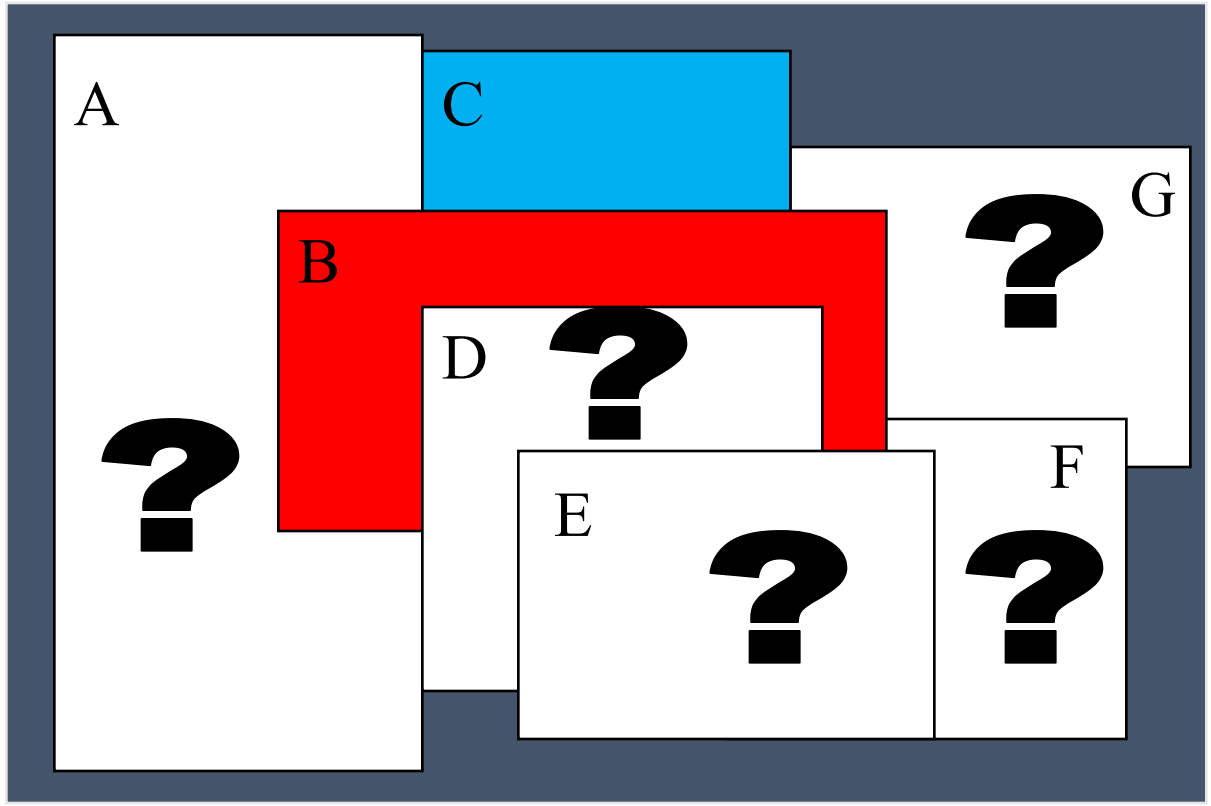
- What is a CSP ?
- CSP is a problem defined through a set of variables, a set of domains and constraints
  - variables can have possible values specified by the problem
  - domains specify the possible values for each variable
  - constraints describe allowable combinations of values for a subset of the variables
- *State* in a CSP
  - defined by an assignment of values to some or all variables
  - *Consistent assignment*: assignment does not violate the constraints
  - An assignment is *complete* when every value is mentioned
- *Solution* to a CSP
  - A *solution* to a CSP is a complete assignment that satisfies all constraints.
    - must assign values to all variables
    - must satisfy all constraints

CSP Approach

- The goal test is decomposed into a set of constraints on variables
  - checks for violation of constraints before new nodes are generated
    - must backtrack if constraints are violated
  - forward-checking looks ahead to detect unsolvability
    - based on the current values of constraint variables

CSP Example: Map Coloring

- Color a map with three colors so that adjacent countries have different colors



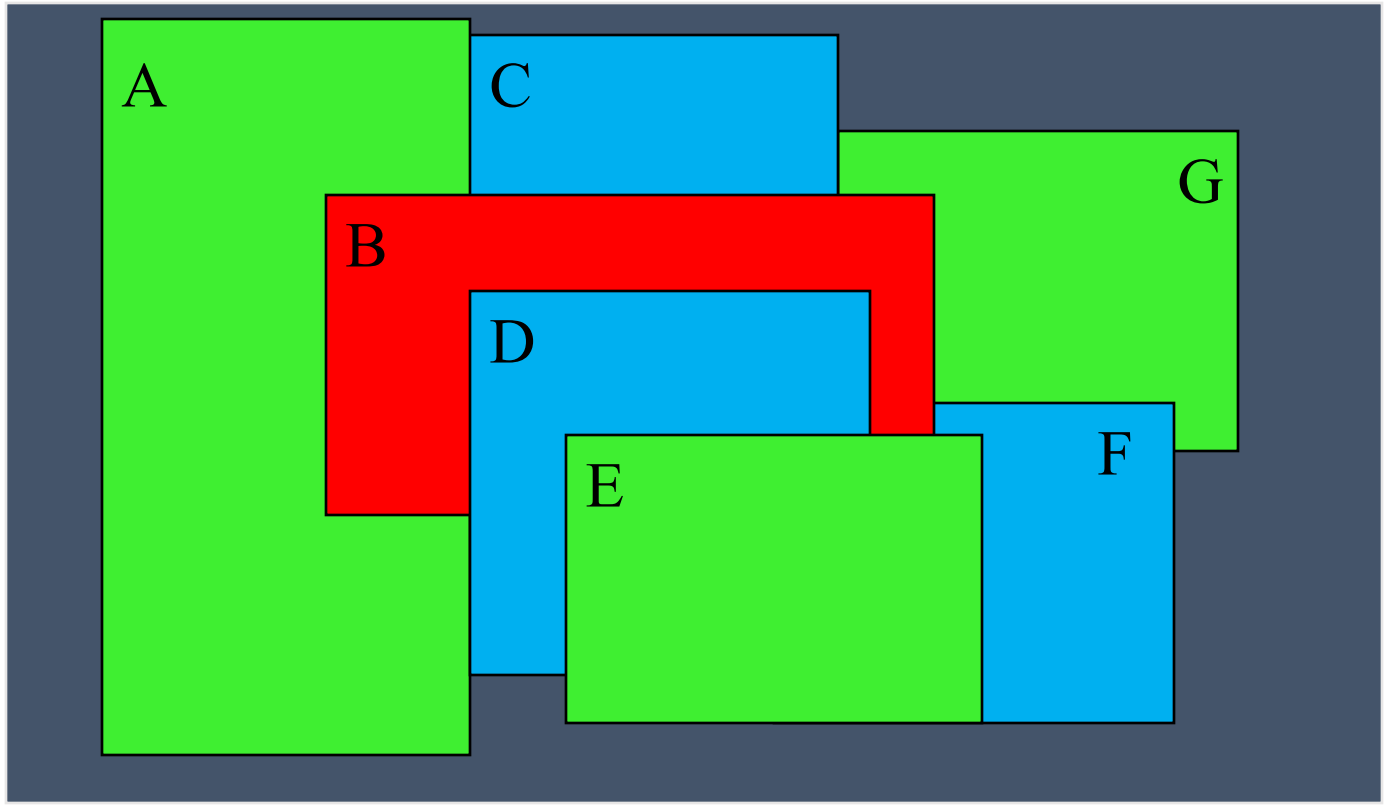
variables:
A, B, C, D, E, F, G

values:
{red, green, blue}

constraints:
"no neighboring regions have the same color"

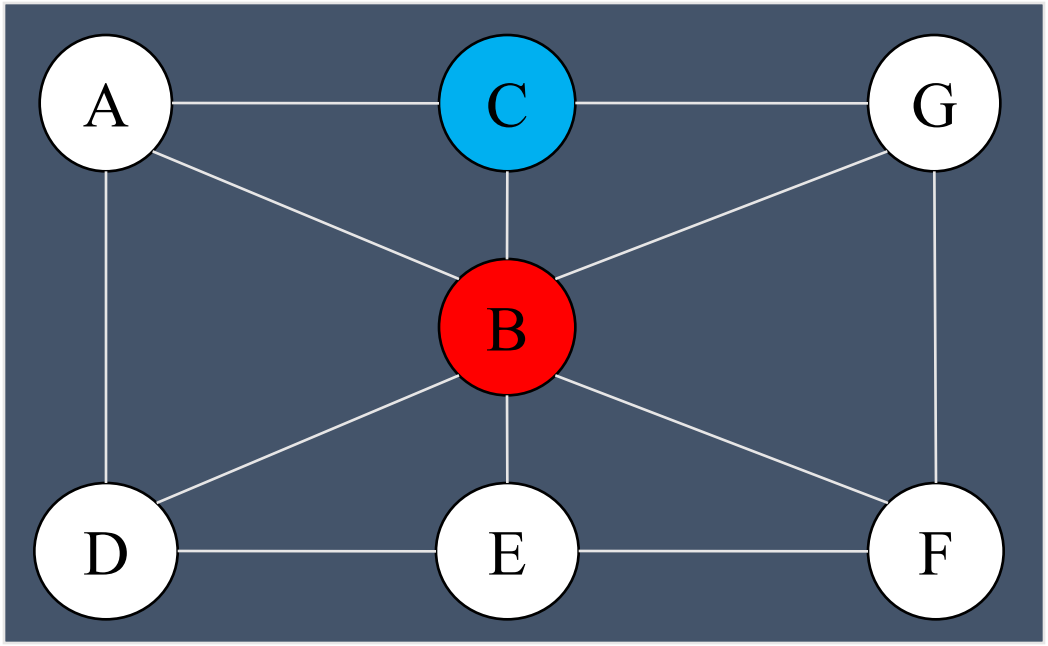
CSP Example: Map Coloring

- Solutions are assignments satisfying all constraints,
- e.g. *{A=green, B=red, C=blue, D=blue, E=green, F=blue, G=green}*



Constraint Graph

- Visual representation of a CSP
  - variables are nodes
  - arcs are constraints



The map coloring example represented as constraint graph. The graph can be used to simplify the search

Varieties of CSPs

- Discrete variables
- These are variables that can take on separate distinct values
  - Finite domains
  - The variables take on only limited set of values
    - E.g. Boolean CSPs, include Boolean satisfiability (NP-complete).
  - Infinite domains (integers, strings, etc.)
  - The variables can take values from an unlimited set
    - E.g. job scheduling, variables are start/end days for each job
    - Need a constraint language e.g $StartJob_1 + 5 \leq StartJob_3$.
    - Linear constraints solvable, nonlinear undecidable.
- Continuous variables
- variables take values from real number ranges
  - e.g. start/end times for Hubble Telescope observations.
  - Linear constraints solvable in poly time by LP methods.

Varieties of constraints

- Unary constraints involve a single variable.
  - e.g. $SA \neq green$
- Binary constraints involve pairs of variables.
  - e.g. $SA \neq WA$
- Higher-order constraints involve 3 or more variables.
  - e.g. cryptharithmetic column constraints.
- Preference (soft constraints) e.g. *red* is better than *green* often representable by a cost for each variable assignment $\rightarrow$ constrained optimization problems.

Backtracking Search Algorithm for CSPs

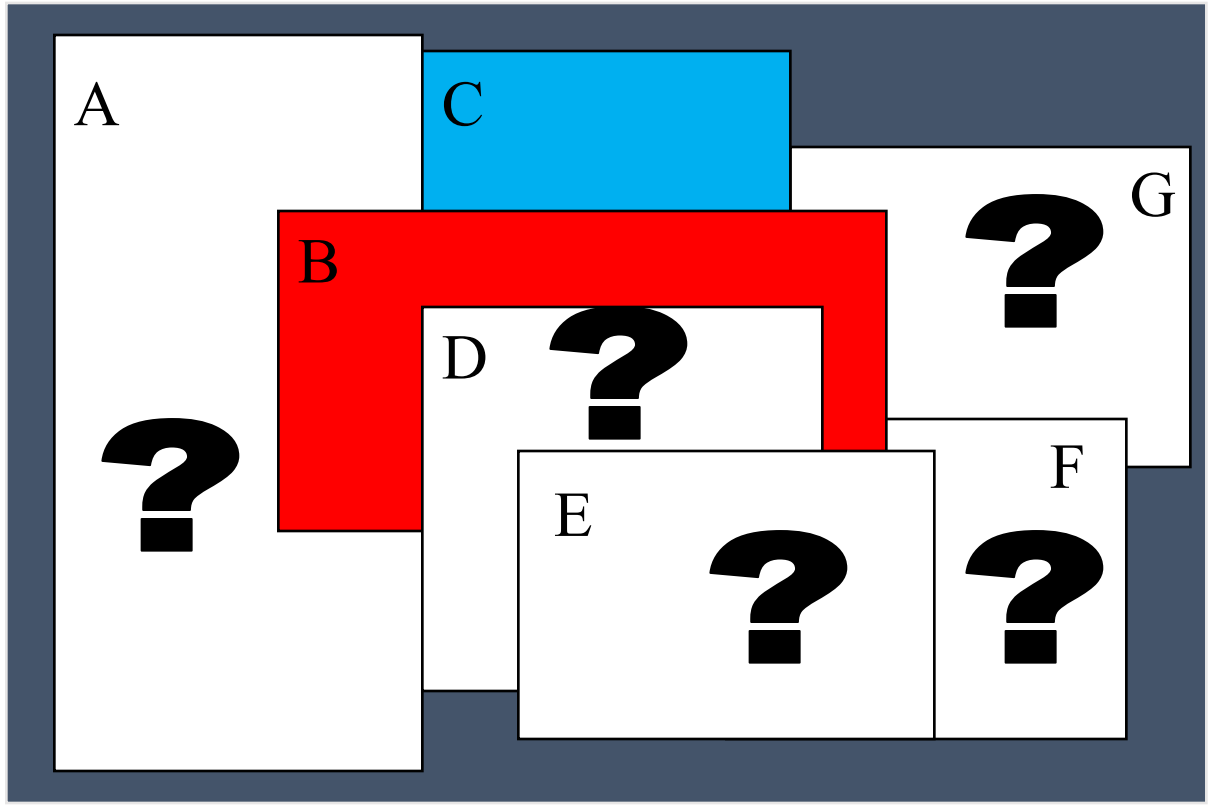
- A variation of depth-first search that is often used for CSPs
- Works likes the basic DFS but with constraints checks
  - values are chosen for one variable at a time
  - if no legal values are left, the algorithm backs up and changes a previous assignment
  - very easy to implement
    - initial state, successor function, and goal test are standardized
  - It is complete and sound
  - not very efficient
    - can be improved by trying to select more suitable unassigned variables first

Enhancements to Backtracking Algorithm

- Heuristics for CSP
- Most-constrained variable (minimum remaining values)
  - variable with the fewest possible values left is selected,
    - i.e. the variable that is most constrained
  - tends to minimize the branching factor
- Most-constraining variable
  - variable with the largest number of constraints on other unassigned variables
  - i.e. the variable that imposes the most constraints on other variables
- Least-constraining value
  - for a selected variable, choose the value that leaves more freedom for future choices

CSP Example: Map Coloring (revisited)

- Color a map with three colors so that adjacent countries have different colors



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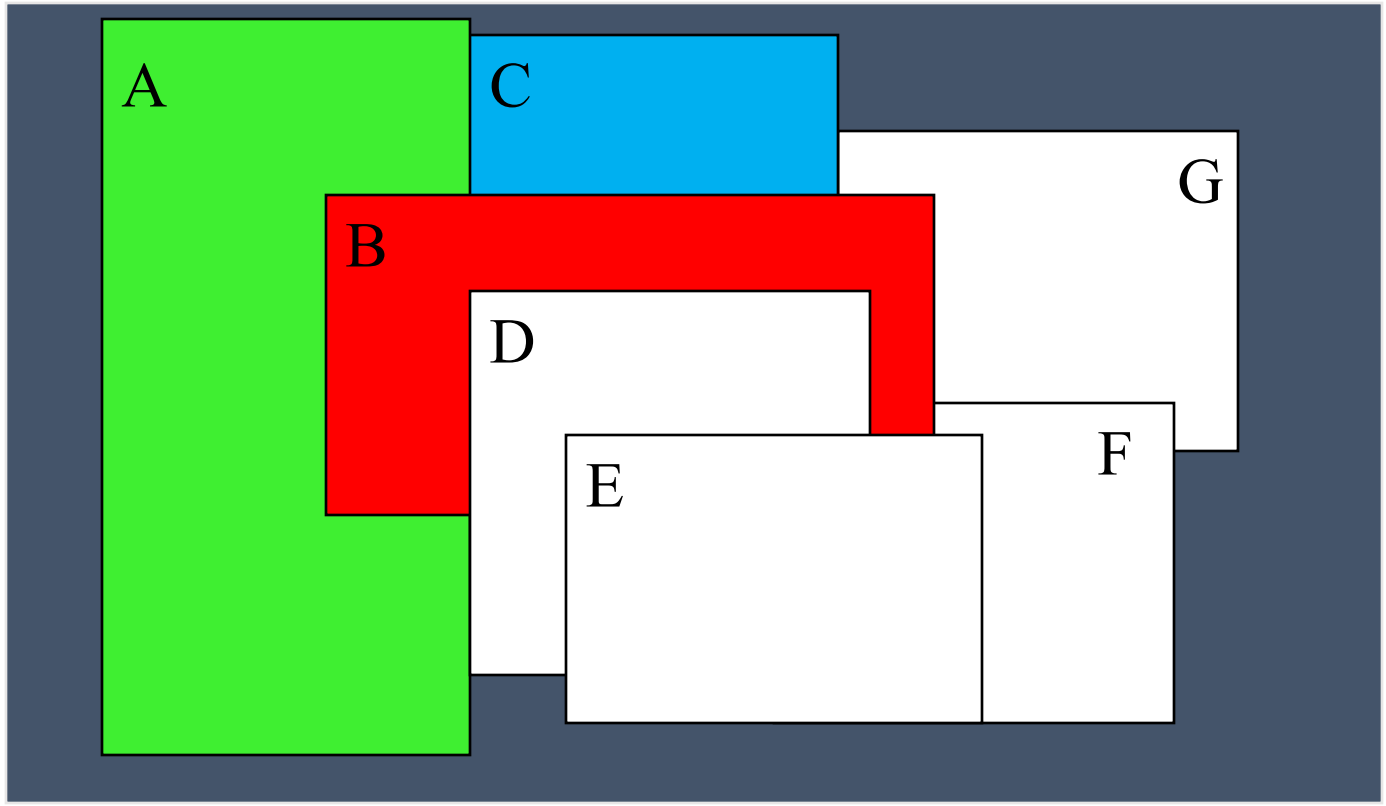
CSP Example: Map Coloring (cont.)

- Most-constrained-variable heuristic



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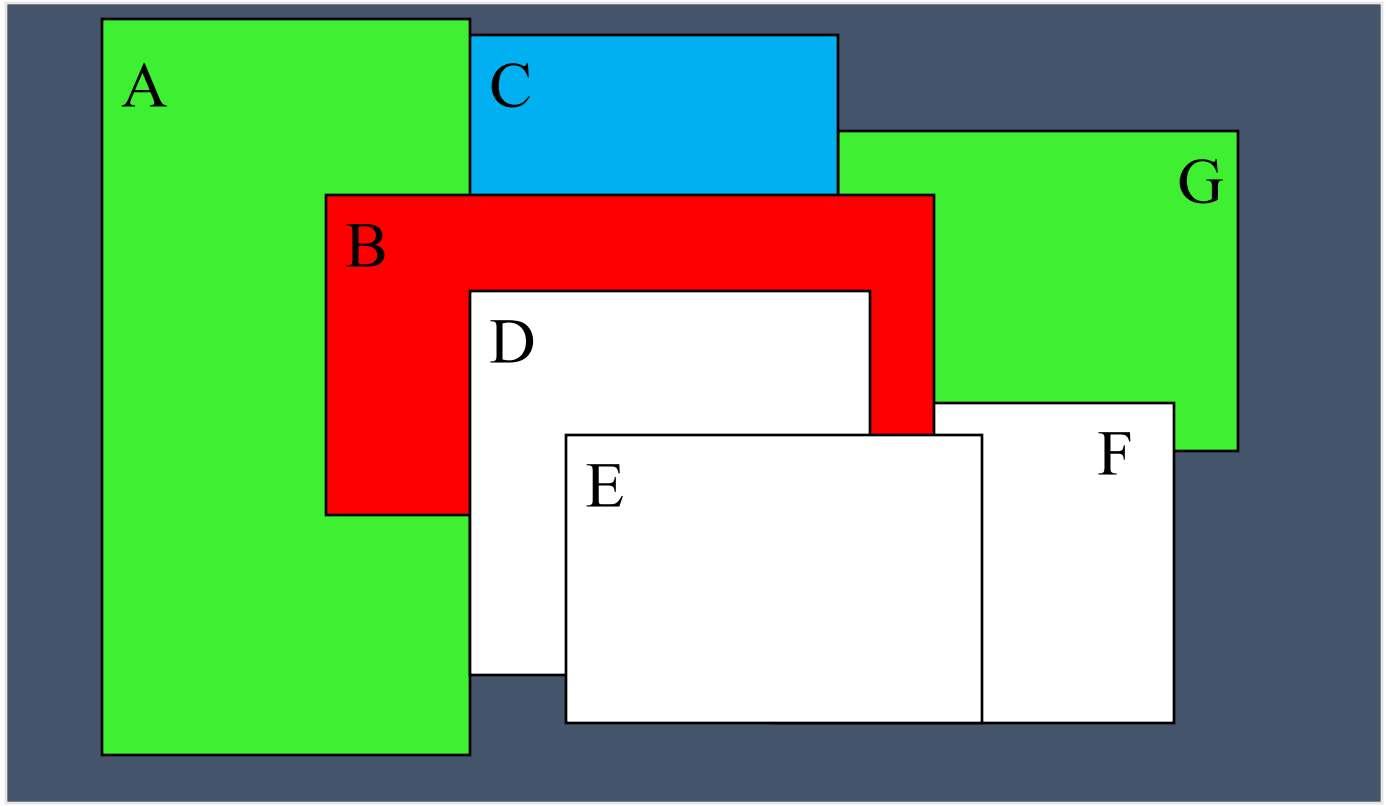

CSP Example: Map Coloring (cont.)

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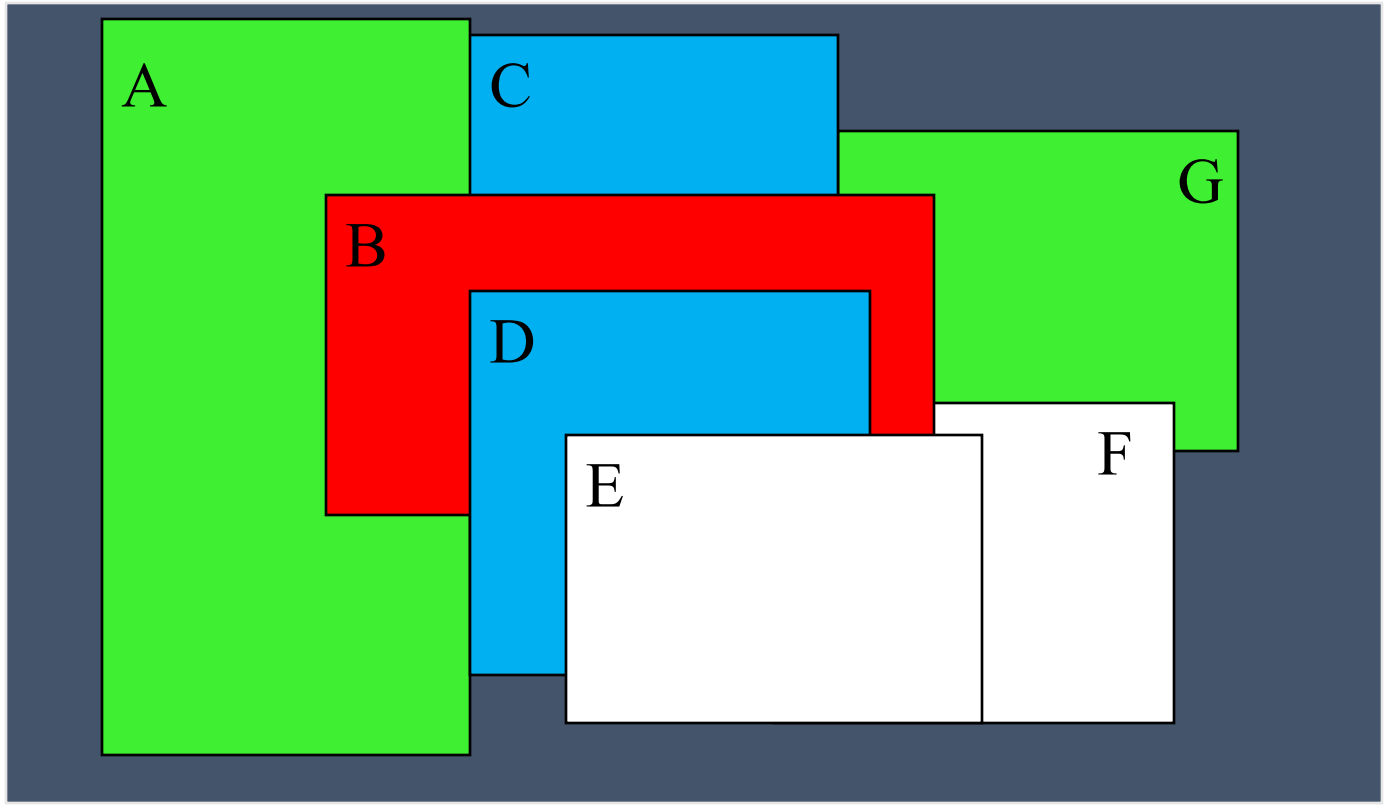

CSP Example: Map Coloring (cont.)

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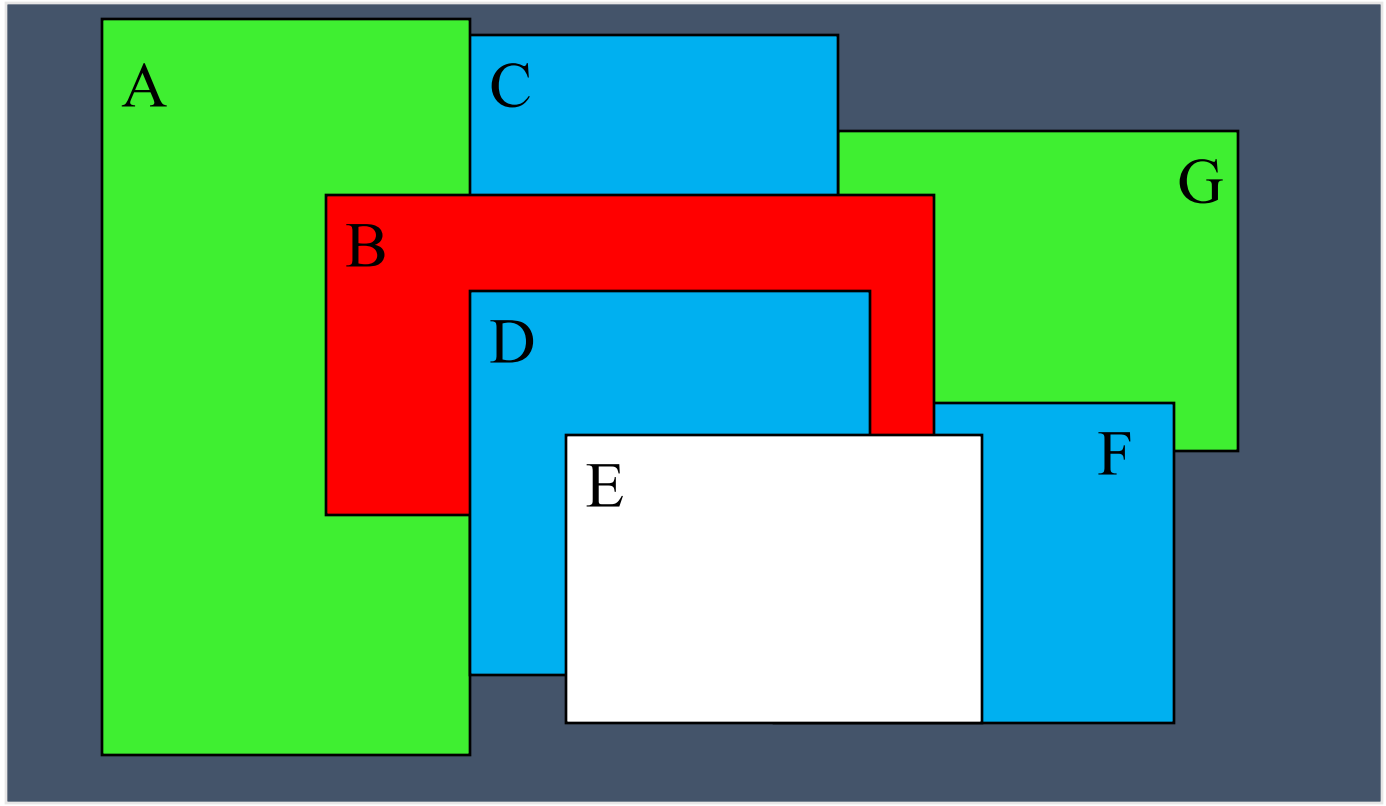

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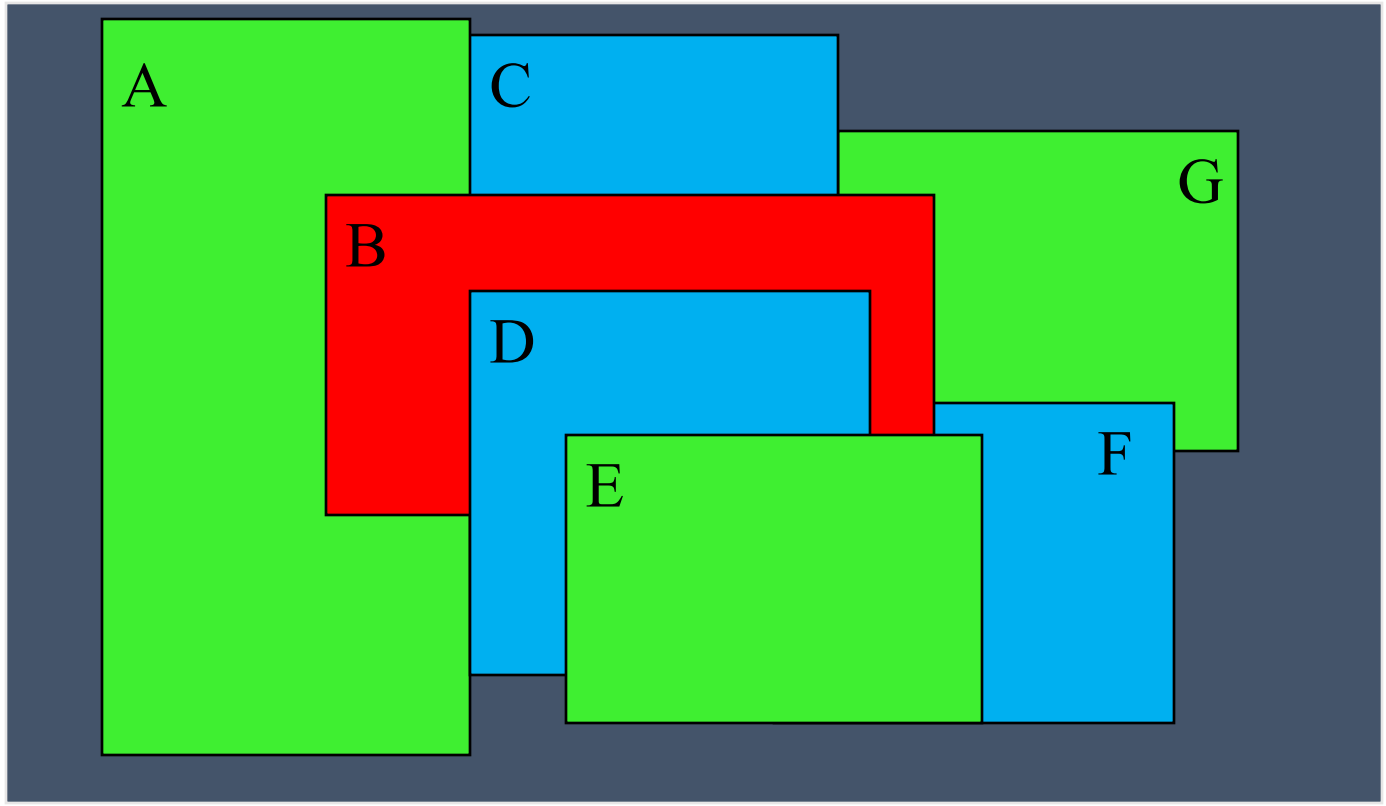
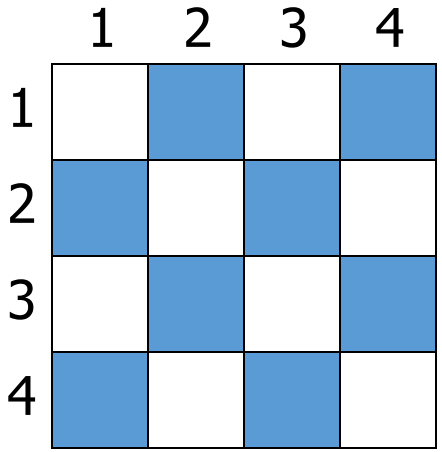
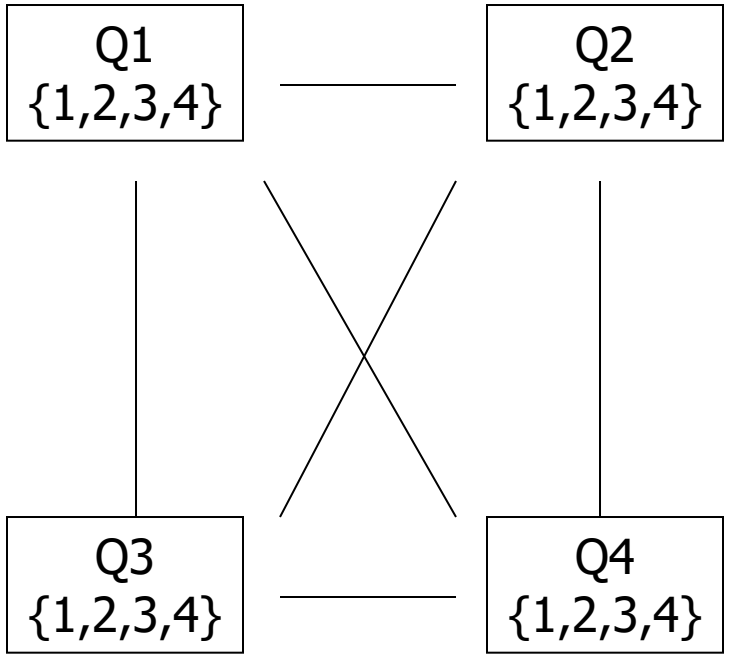


Enhancements to Backtracking Algorithm

- Forward checking
- Can we detect inevitable failure early?
  - *And avoid it later?*
- when a value *X* is assigned to a variable, inconsistent values are eliminated for all variables connected to *X*
    - identifies “dead” branches of the tree before they are visited
- *Forward checking idea:* keep track of remaining legal values for unassigned variables.
- Terminate search when any variable has no legal values

Example: 4-Queens Problem





[4-Queens slides copied from B.J. Dorr CMSC 421 course on AI]

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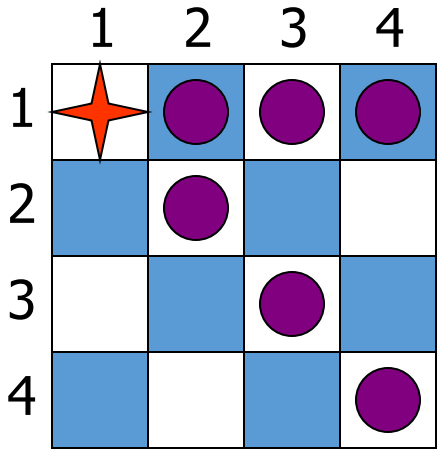

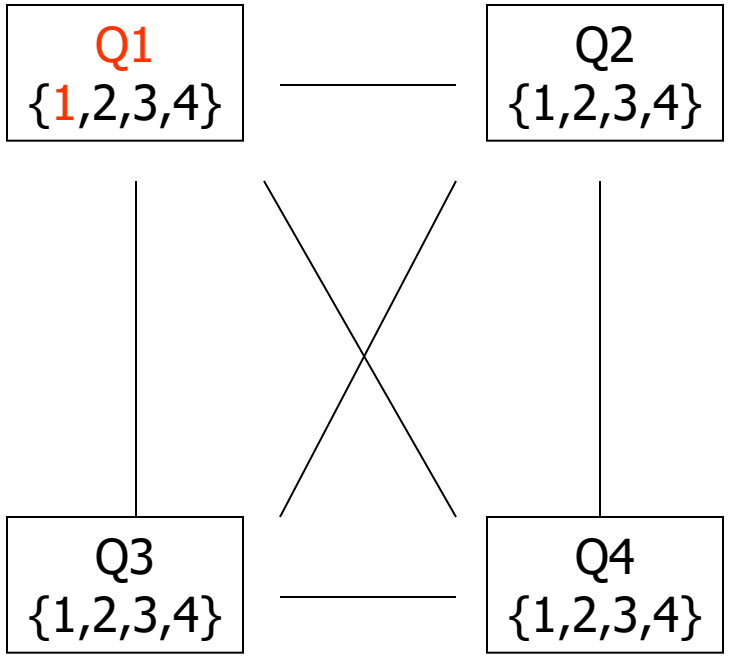

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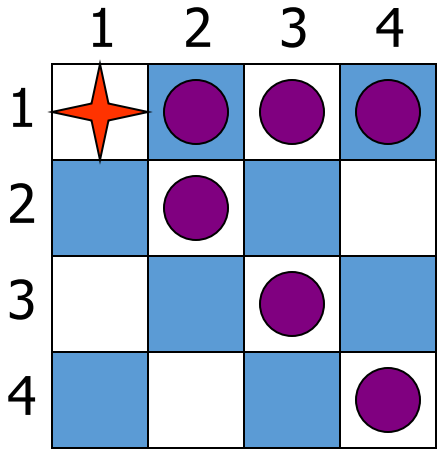

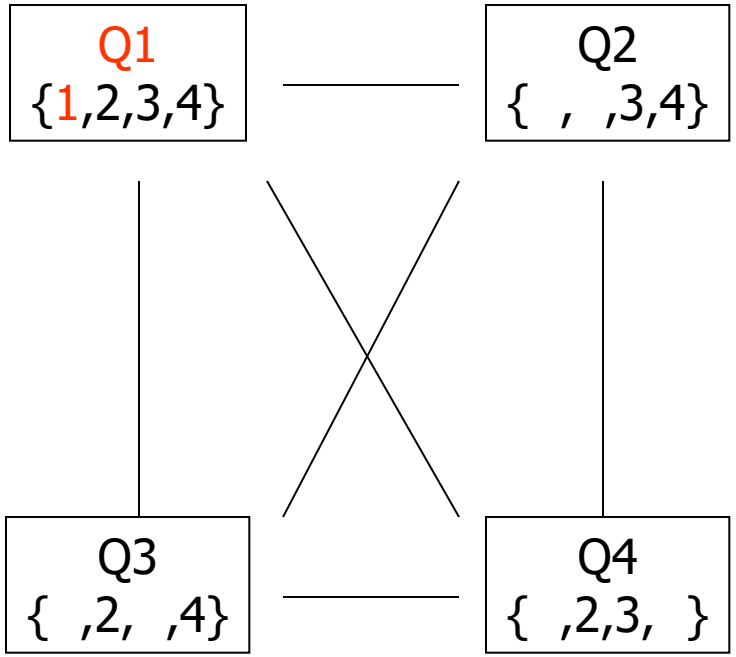

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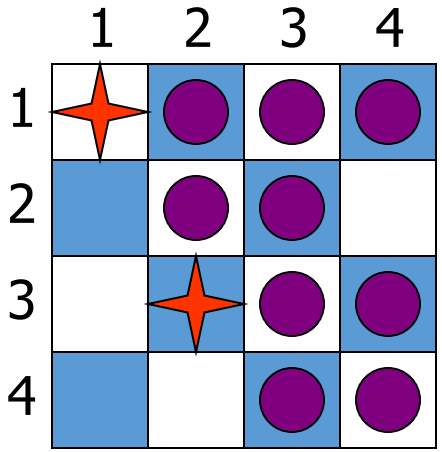

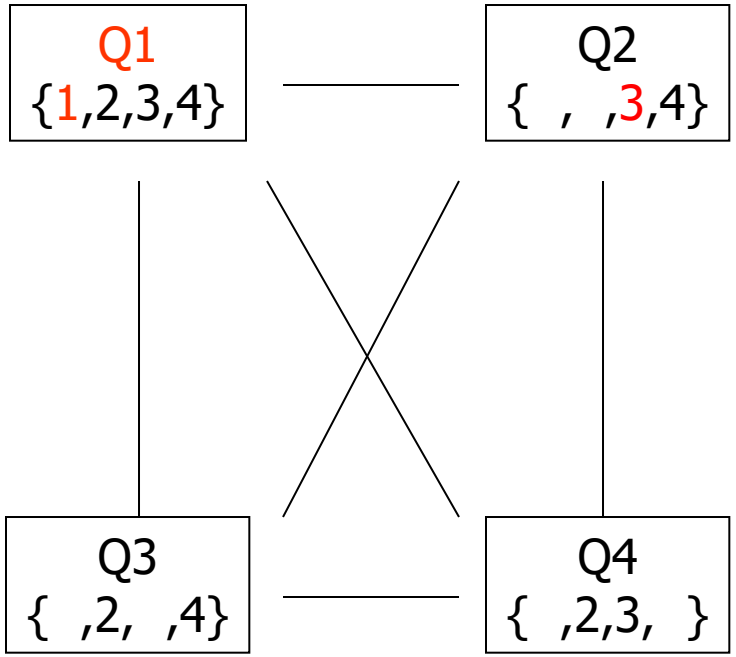

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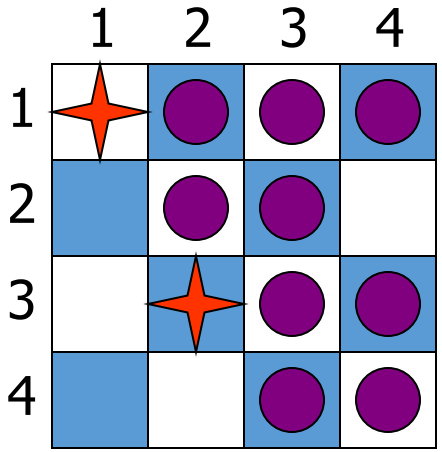

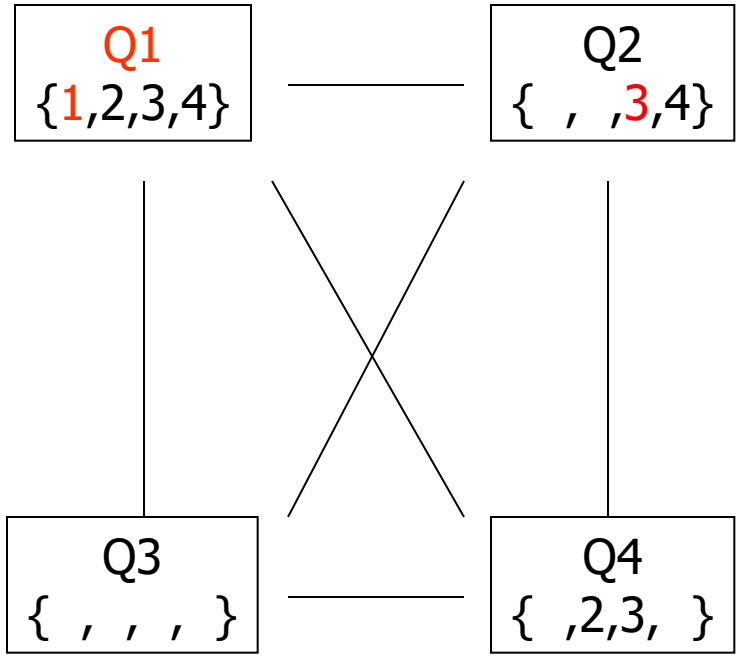

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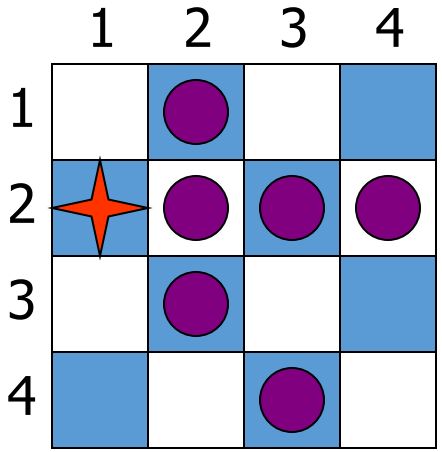

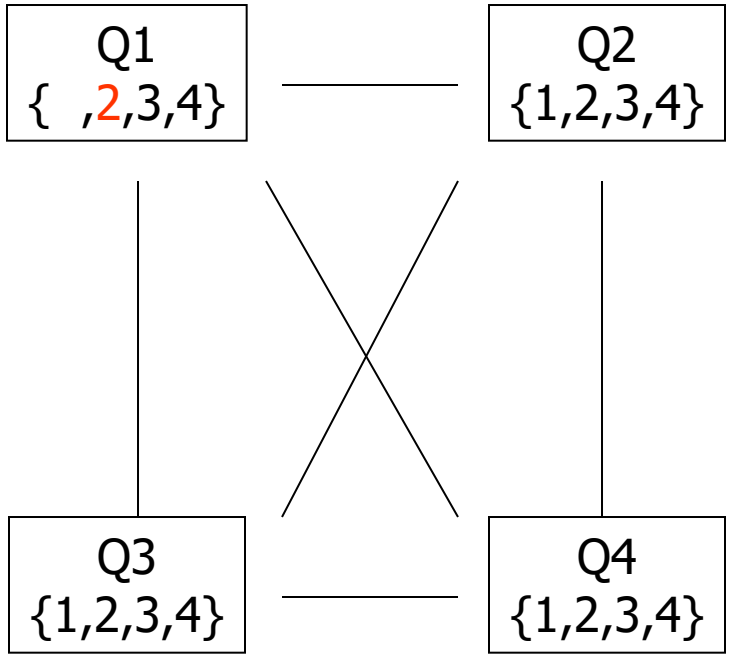

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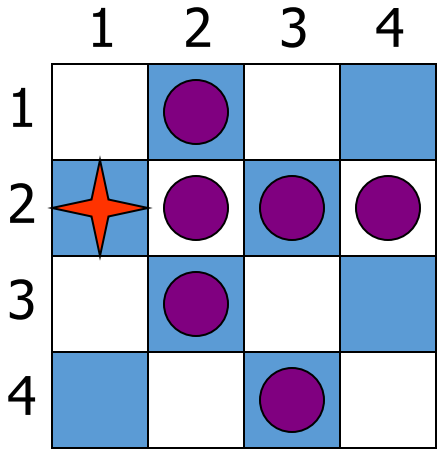

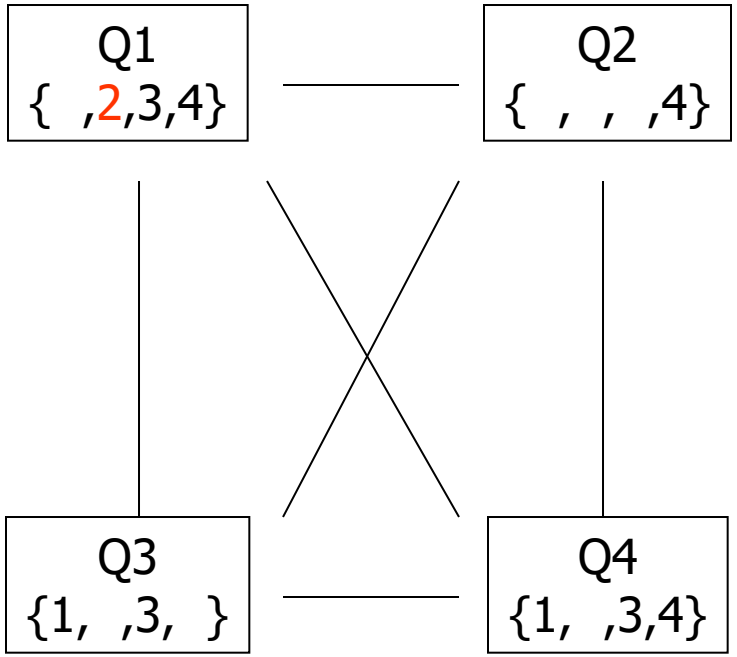

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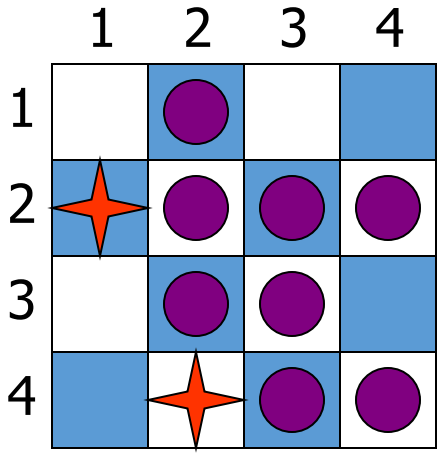

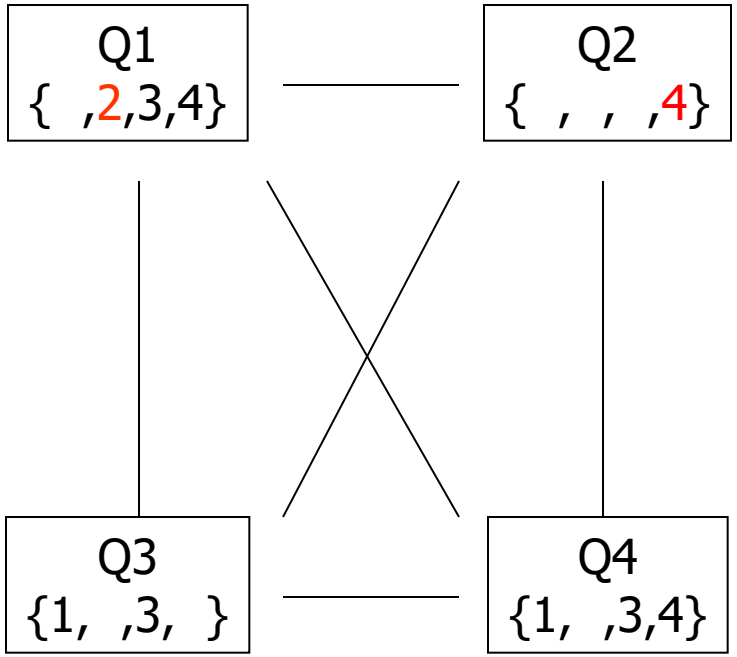

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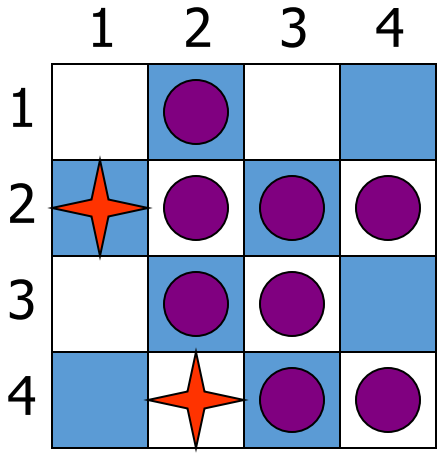

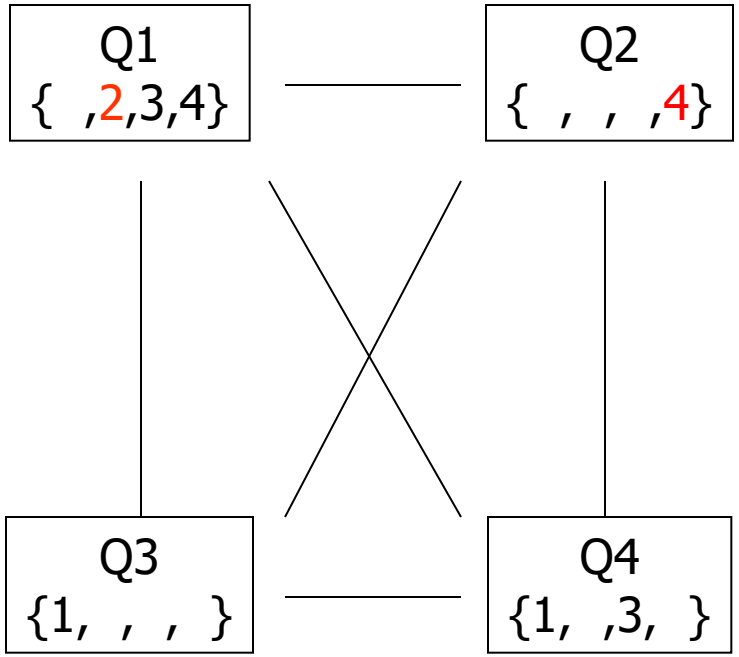

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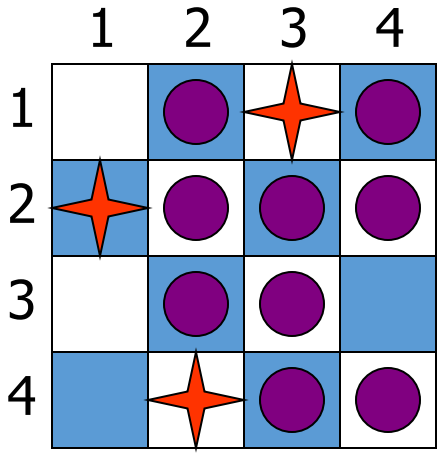

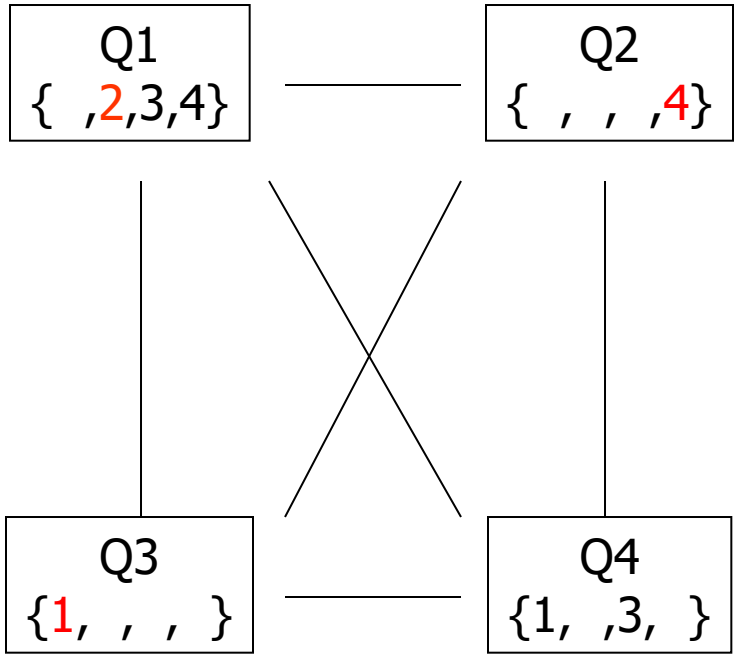

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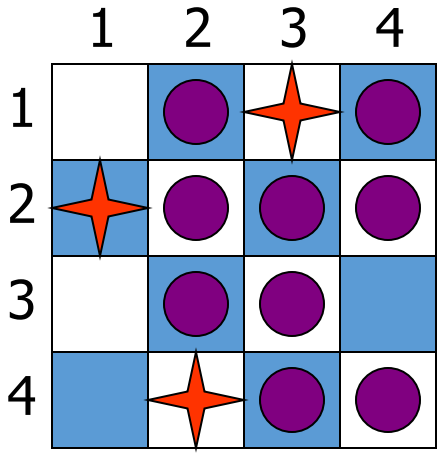

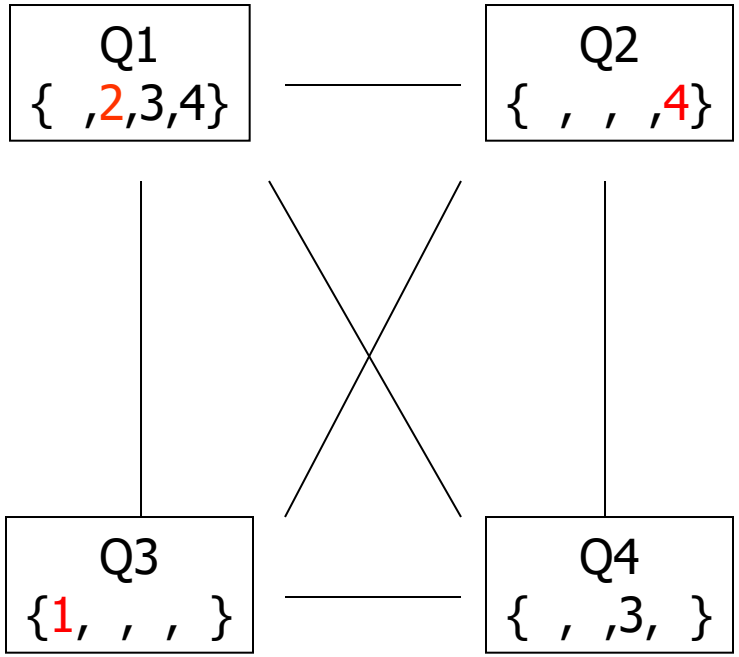

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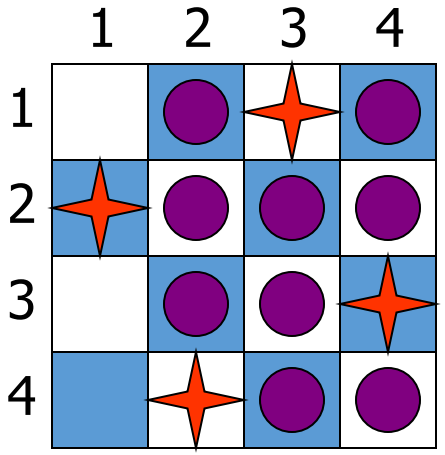
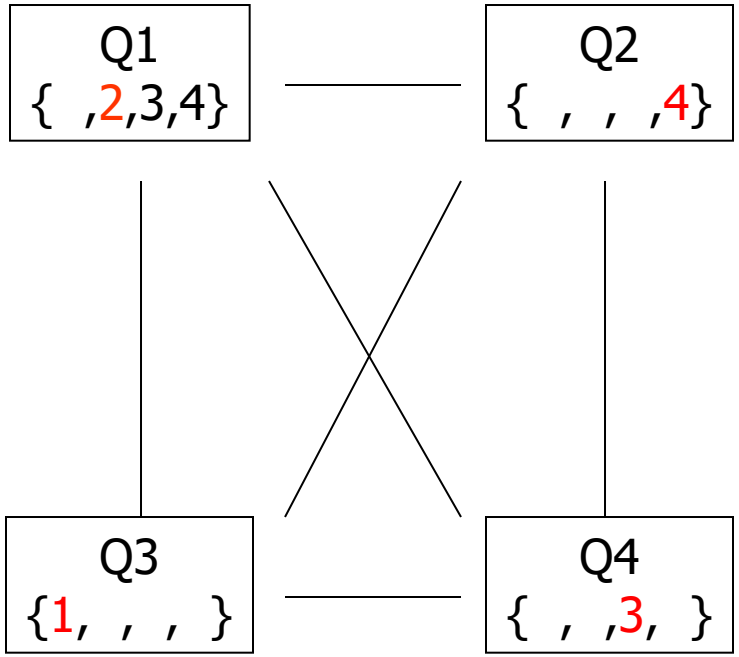




Local search for CSPs

- Note: The path to the solution is unimportant, so we can apply local search!

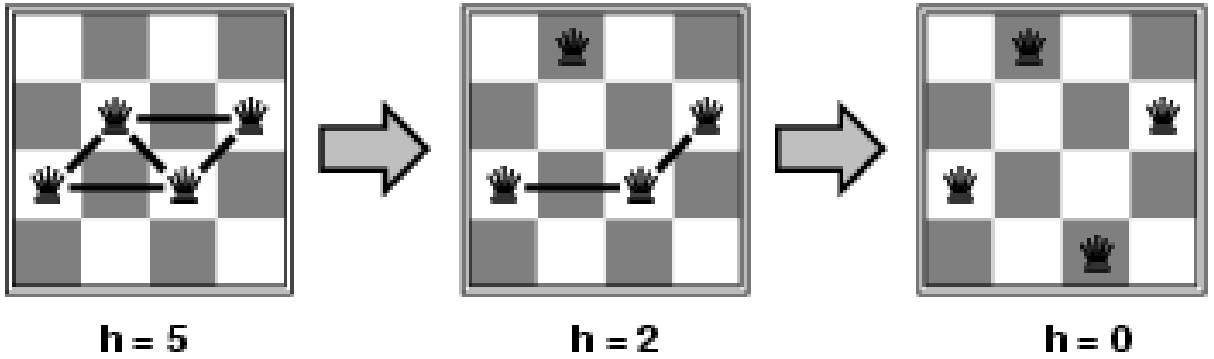
- To apply to CSPs:
  - allow states with unsatisfied constraints
  - operators reassign variable values

- Variable selection: randomly select any conflicted variable

- Value selection by min-conflicts heuristic:
  - choose value that violates the fewest constraints
  - i.e., hill-climb with *h(n)* = total number of violated constraints

Example: 4-Queens

- States: 4 queens in 4 columns (4^4 = 256 states)
- Actions: move queen in column
- Goal test: no attacks
- Evaluation: *h(n)* = number of attacks



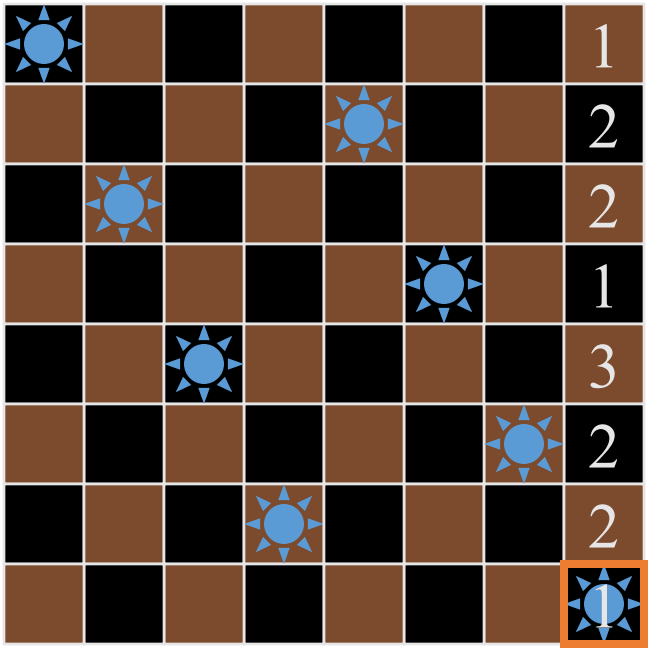
Local Search and CSP

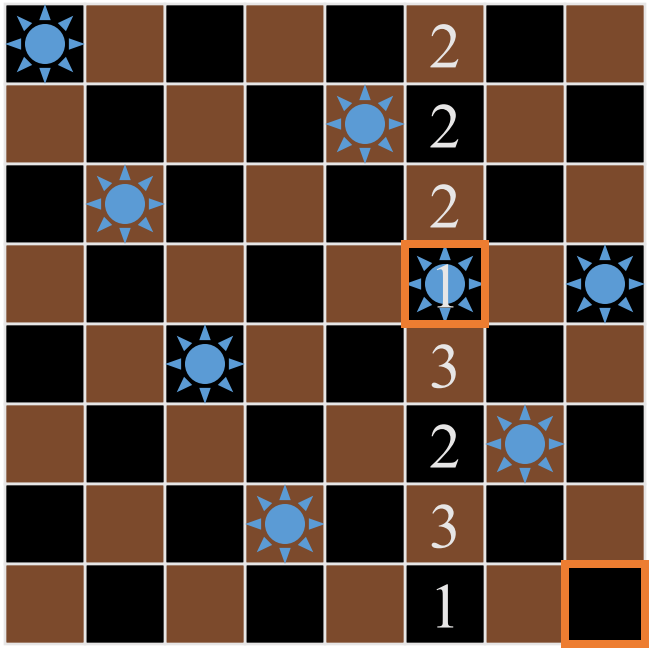
- Local search (iterative improvement) is frequently used for constraint satisfaction problems
  - values are assigned to all variables
  - modification operators move the configuration towards a solution
- Often called heuristic repair methods
  - repair inconsistencies in the current configuration
- Simple strategy: min-conflicts
  - minimizes the number of conflicts with other variables
  - solves many problems very quickly

8-Queens with Min-Conflicts

- One queen in each column
  - usually several conflicts
- Calculate the number of conflicts for each possible position of a selected queen
- Move the queen to the position with the least conflicts

8-Queens Example





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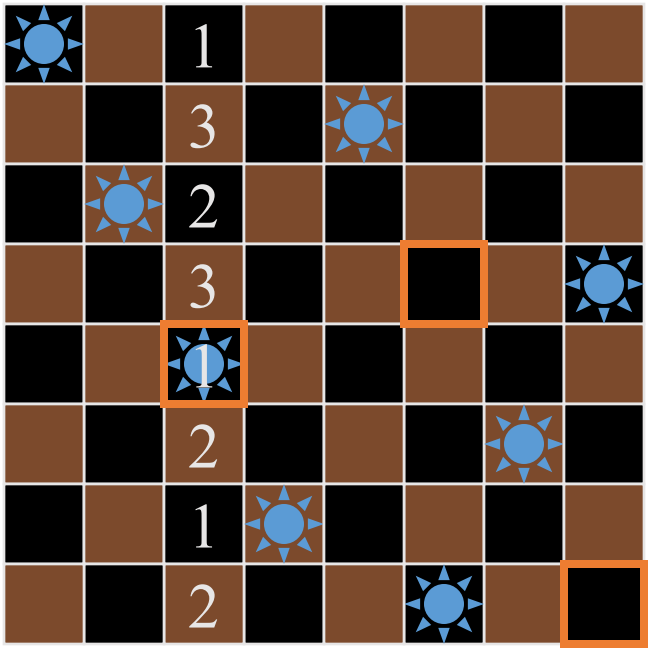

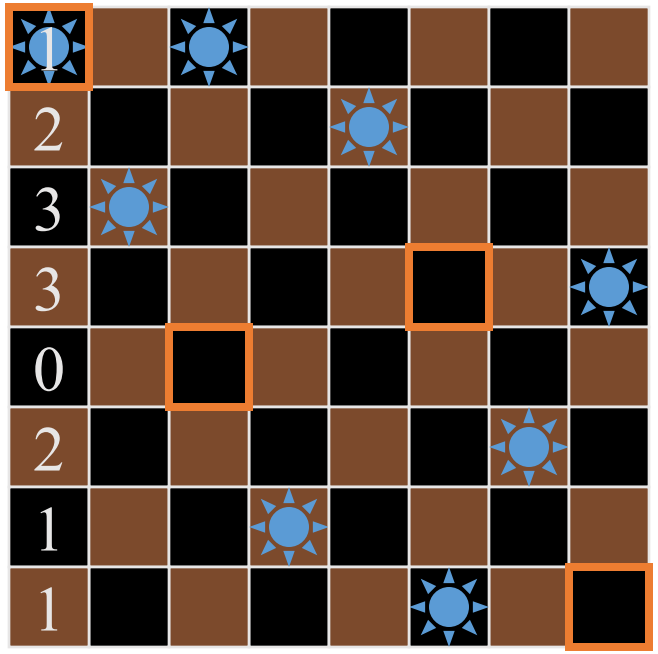

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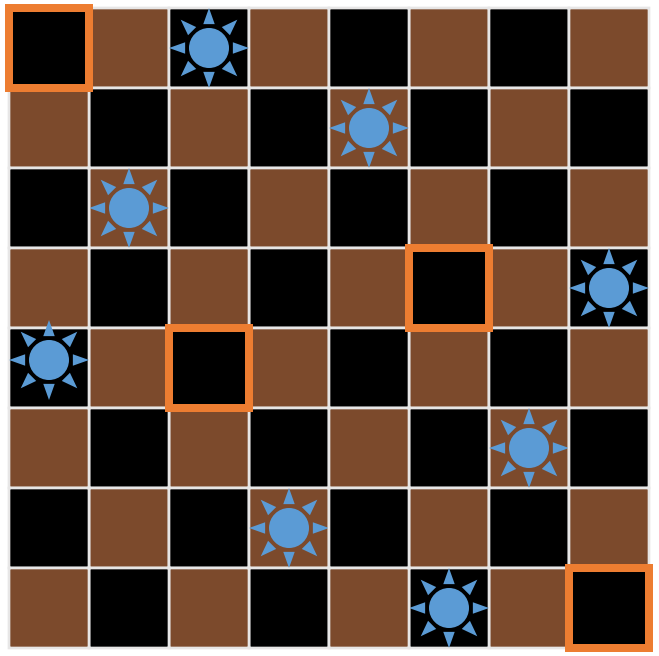


- Solution found in 4 steps
- Min-conflicts heuristic
- Uses additional heuristics to select the “best” queen to move
  - try to move out of the corners
    - similar to least-constraining value heuristics

Advantages of local search

- The runtime of min-conflicts is roughly independent of problem size.
  - Solving the millions-queen problem in roughly 50 steps.
- Local search can be used in an online setting.
  - Backtrack search requires more time

Summary

- CSPs are a special kind of problem:
  - states defined by values of a fixed set of variables
  - goal test defined by constraints on variable values
- Backtracking = depth-first search with one variable assigned per node
- Variable ordering and value selection heuristics help significantly
- Forward checking prevents assignments that guarantee later failure
- Iterative min-conflicts is usually effective in practice

Week 3: Review Questions

- 1. What is a Constraint Satisfaction Problem (CSP)?
- ii. Define the three main components of a CSP.
- iii. Provide an example of a real-world problem that can be modeled as a CSP.
- 2. Explain the terms "consistent assignment" and "complete assignment" in the context of CSPs.
- ii. How do these terms relate to finding a solution for a CSP?
- 3. Describe the Constraint Graph.
- ii. How does it help in solving CSPs?
- Iii. Draw the constraint graph for the map coloring example with regions A, B, C, D, E, F, and G.
- 4. What are the differences between discrete and continuous variables in CSPs?
- ii. Give an example of a problem for each type.
- 5. Explain the Most-Constrained Variable and Least-Constraining Value heuristics.
- ii. How do these heuristics improve the efficiency of backtracking search?
- 6. What is Forward Checking in backtracking search?
- ii. How does it help in detecting dead ends early?

Week 3: Review Questions

- 7. How does the Min-Conflicts heuristic work in local search for CSPs?

  ii. Why is it particularly effective for large problems like the N-Queens problem?
- 8. Compare and contrast backtracking search and local search for solving CSPs.

  ii. What are the advantages and disadvantages of each method?
- 9. Solve the 6-Queens problem using:

  i. Backtracking search with the Most-Constrained Variable heuristic.

  ii. Local search with the Min-Conflicts heuristic.
- Requirements:

  a. For backtracking, show the search tree up to the first solution.

  b. For local search, start with a random configuration and show each step until a solution is found.

  c. Compare the efficiency of the two methods for this problem.
-

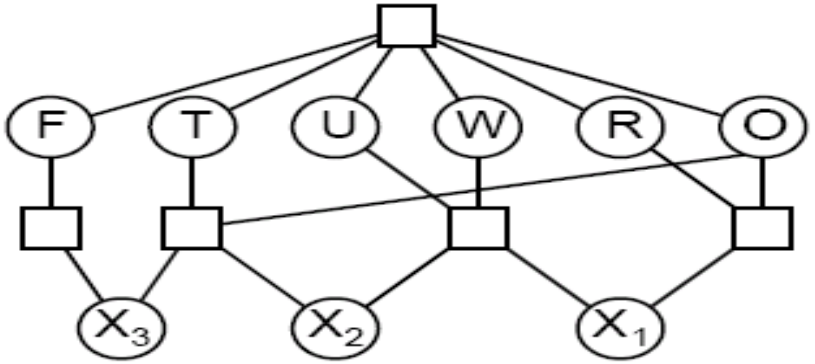
Week 3: Review Questions

- 10. Model one of the following real-world problems as a CSP:
  i. University class scheduling (courses, rooms, times, professors).
  ii. Sudoku puzzle.
  Iii. Resource allocation in a factory.
- Requirements:
  a. Define the variables, domains, and constraints.
  b. Explain how you would solve the problem using either backtracking or local search.
  c. Discuss potential challenges and how heuristics could help.

Week 3: Review Questions

- 11.

$$\begin{array}{r} T\ W\ O \\ +\ T\ W\ O \\ \hline F\ O\ U\ R \end{array}$$



Variables: $F\ T\ U\ W\ R\ O\ X_1\ X_2\ X_3$
Domains: $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
Constraints
alldiff(F, T, U, W, R, O)
$O + O = R + 10 \cdot X_1$, etc.